# Local Circuit Rewriting, Made Exact

Equivalence as a group action, certified ground truth, and a hardware-aware optimiser

Draft extension of Rosenhahn, Osborne & Hirche, arXiv:2502.14715

**Abstract**

Local term-replacement methods shorten a quantum circuit by repeatedly replacing a short sub-circuit with a cheaper equivalent one from a precomputed database. Rosenhahn, Osborne and Hirche [1] recently proposed such a method and compared it with Qiskit. We make three contributions. (i) *A framework.* Every useful notion of "equivalent sub-circuit" (exact, up to global phase, on the $|0\rangle$ input, before a computational-basis measurement, up to a symmetry of the chip, within $\varepsilon$) is the orbit relation of a group action on unitaries. The database is then keyed by a canonical form of the orbit, and soundness and error accumulation follow from two short propositions. (ii) *A general optimiser.* It makes no assumption about the gate set. It uses hashed databases of up to $3.2 \times 10^7$ three-qubit classes, meet-in-the-middle lookup, exact two-qubit resynthesis, and DAG-convex windows restricted to coupled qubits. It optimises a calibrated cost ($-\log$ fidelity per gate), which we select over gate count and two-qubit-weighted count by noisy simulation. (iii) *Exact ground truth.* Every gate in [1] is Clifford, so its benchmarks live in a finite group that we enumerate completely (92,897,280 three-qubit elements). This gives proven optima, an exactness theorem for local rewriting, a parity theorem, and a proof that random-circuit benchmarks converge to a known constant. Against this ground truth, the general optimiser (which does not use it) reaches the proven optimum on 17/20 and 18/20 three-qubit benchmark circuits and produces 0.25× and 0.42× the length of the original method's output. On 58 algorithm circuits (Nam/VOQC, QASMBench, QFT/QAOA/Trotter) compiled for an IBM Heron model, it removes 13% of the gates of Qiskit level-3 output without ever adding a two-qubit gate. Its output has a lower noisy-simulation error than Qiskit level 3 on 39/49 circuits. All 250 paper-benchmark outputs and all 720 hardware-suite outputs are verified numerically.

## 1 Introduction

Compiling a circuit into a device's native gates leaves redundancy, and on noisy hardware every gate costs fidelity. Rosenhahn, Osborne and Hirche [1] proposed a simple, gate-set-agnostic remedy: view the circuit as a word over the native alphabet, sample a window of 2–8 letters, and replace it by a shorter equivalent word, found by random sampling (V1), by lookup in an exhaustively enumerated database of short circuits (V2), or by a lookup gated by a random forest (V3). The method belongs to a long line of database-driven and peephole optimisers [2, 3, 4, 5, 6, 7, 8, 9].

Two questions decide whether such a method is worth using, and [1] answers neither. *How far from optimal is it?* The paper compares only with Qiskit, so it cannot say whether its 45-gate output for a 3-qubit circuit is good (the optimum is about 10). *Does it help on circuits and hardware that matter?* The paper's alphabets are discrete $\pm\pi/2$ rotations on all-to-all connectivity. Its benchmarks are random words over those alphabets, and its objective is the total gate count.

This paper answers both questions. The answers rest on two pieces that we keep strictly separate:

- a **general optimiser** (§2–3) that uses no structure of the gate set. It runs on continuous-angle native gates, respects connectivity, and supports exact, state-aware and approximate equivalence;
- an **exact ground truth** (§4–5) that *does* use structure. All gates of [1] are Clifford, so their circuits are elements of a finite group whose Cayley graph we enumerate completely. We use it only to evaluate; the optimiser never sees it.

**Contributions.**

1. A unifying model: equivalence notions are group actions, and database keys are canonical orbit representatives (Table 1). Soundness of state- and measurement-aware rewriting and an additive error bound for approximate rewriting follow from this model (Props. 1, 2).
2. A general, connectivity-aware optimiser with exact global-phase handling and a calibrated cost model (§3).
3. Theory for the discrete case: an exactness threshold (local = global optimality), a parity theorem, mixing of random-circuit benchmarks, and universal counting lower bounds (§4).
4. Exact ground truth for every 3-qubit circuit over both alphabets of [1], and certified bounds for 4 qubits (§5).
5. Experiments: optimality gaps on the paper's own benchmarks, a hardware suite on IBM Heron and trapped-ion models against Qiskit and TKET, selection of the objective by noisy simulation, and every output verified (§6).
6. An explicit list of the original paper's weaknesses and of the limitations of ours (§7, §8).

# 2 Framework

## 2.1 Circuits, devices, costs

A *device* fixes $n$ physical qubits, a coupling graph $E$ (or all-to-all), native single-qubit gates, a native two-qubit gate allowed only on pairs in $E$, and which angles are continuous. We use two models. **IBM Heron** (Qiskit's `FakeTorino`, 133 qubits, heavy-hex): $\mathrm{RZ}(\theta)$ (virtual), SX, $X$, CZ, with per-qubit and per-pair calibrated errors. Eleven couplers have calibrated error 1.0 and are removed. **Trapped ion** (all-to-all): $\mathrm{RX}, \mathrm{RY}, \mathrm{RZ}(\theta)$ and the Mølmer–Sørensen gate $\mathrm{RXX}(\theta)$ at any angle. No public calibration exists, so we use model errors $3 \cdot 10^{-5}$ (1q), $10^{-3}$ (2q), and treat RZ as virtual. A circuit is a word $u = \gamma_1 \cdots \gamma_L$ of native gates with $\mathrm{ev}(u) = \gamma_L \cdots \gamma_1$. A *cost* is additive, $c(u) = \sum_i c(\gamma_i)$. We compare

$$c_{\mathrm{count}} \equiv 1, \qquad c_{w2} = \begin{cases} 10 & \text{2q} \\ 1 & \text{1q} \end{cases}, \qquad c_{\mathrm{cal}}(\gamma) = -\log\bigl(1 - e(\gamma)\bigr),$$

where $e(\gamma)$ is the calibrated error of $\gamma$ on its physical qubits. Then $\exp(-c_{\mathrm{cal}}(u))$ is the estimated success probability (ESP) [13]: minimising $c_{\mathrm{cal}}$ maximises ESP. Virtual gates get a tie-breaking cost of $10^{-7}$.

## 2.2 Equivalence as a group action

A replacement $u \to v$ of a window is *sound* if it does not change what the full program computes. What counts as "the same" depends on context, and each notion we need is the orbit relation of a group $H$ acting on $U(2^k)$:

| notion | $U \sim V$ iff | group acting | canonical key |
|---|---|---|---|
| exact | $V = U$ | trivial | $U$ |
| up to phase (as in [1]) | $V = e^{i\varphi}U$ | $U(1)$, left | phase-normalised $U$ |
| input-aware (at circuit start) | $V\lvert 0\rangle = e^{i\varphi}U\lvert 0\rangle$ | stabiliser of $\lvert 0\rangle$; $U(1)$ | phase-normalised $U\lvert 0\rangle$ |
| output-aware (before measurement) | $V = DU$, $D$ diagonal | diagonal unitaries, left | rows phase-normalised |
| symmetry | $V = P_\pi U P_\pi^\dagger$ | coupling-graph automorphisms | minimum over the orbit |
| approximate | $\lVert U - V\rVert \le \varepsilon$ | none (a tolerance) | exact key, verified distance |

Table 1: Equivalence notions. The database is keyed by a hash of the canonical form, so one lookup finds every member of the orbit. Notions compose: for example, output-aware equivalence modulo automorphisms uses keys minimised over the automorphism group of row-normalised matrices.

Exact equivalence including global phase is the only notion that stays sound when the circuit

is later controlled or used as a sub-routine. "Up to phase" is what [1] uses. We keep it only to compare with that paper.

**Proposition 1** (Soundness of state- and measurement-aware rewriting)**.** *Let the program be "prepare $|0^n\rangle$, apply $\mathrm{ev}(u)$, measure all qubits in the computational basis". Let $S$ be a convex window of $u$ on wires $W$. (a) If no gate precedes $S$ on any wire in $W$ and $\mathrm{ev}(v)|0_W\rangle = e^{i\varphi}\mathrm{ev}(S)|0_W\rangle$, replacing $S$ by $v$ does not change the output distribution. (b) If no gate follows $S$ on any wire in $W$ and $\mathrm{ev}(v) = D\,\mathrm{ev}(S)$ with $D$ diagonal, replacing $S$ by $v$ does not change the output distribution. Any sequence of such rewrites, each applied to the current circuit, preserves the output distribution.*

*Proof.* By convexity we may move $S$ to the front (a) or the back (b) of the circuit without changing $\mathrm{ev}(u)$. (a) The input factorises as $|0_W\rangle \otimes |0_{\bar{W}}\rangle$, so the final state changes only by $e^{i\varphi}$. (b) The final state changes from $|\psi\rangle$ to $(D \otimes I)|\psi\rangle$, and $|\langle x|(D \otimes I)|\psi\rangle|^2 = |\langle x|\psi\rangle|^2$ for every basis state $x$. The invariant is the output distribution of the current circuit, so soundness composes. □

**Proposition 2** (Approximate rewriting)**.** *If rewrites $i = 1, \ldots, m$ replace windows $S_i$ by $v_i$ with $\|\mathrm{ev}(v_i) - \mathrm{ev}(S_i)\| \le \varepsilon_i$ in operator norm, then the final circuit $u'$ satisfies $\|\mathrm{ev}(u') - \mathrm{ev}(u)\| \le \sum_i \varepsilon_i$. In exact mode this includes global phase.*

*Proof.* Telescope over intermediate circuits. Each step changes one factor of a product of unitaries, and $\|A(X - Y)B\| = \|X - Y\|$ for unitary $A, B$. □

The optimiser tracks $\sum_i \varepsilon_i$ and reports it with the circuit. Equivalence tests are *linear* in the operator error (norm distance, not $|\mathrm{tr}(U^\dagger V)|/2^k \approx 1$, which accepts errors of order $\sqrt{\mathrm{tol}}$).

### 2.3 Local rewriting

**Definition 1** (Local rewriting)**.** With window bound $w$ and database radius $r$, $\mathcal{R}(w, r) = \{u \to v : |u| \le w,\ \mathrm{ev}(u) \sim \mathrm{ev}(v),\ |v| \le r,\ c(v) < c(u)\}$. Two gates are *independent* if they commute exactly. Words equal up to swapping adjacent independent gates are *trace-equivalent* [17, 18]. A word is $(w, r)$-*reduced* if no trace-equivalent word has a factor that is the left side of a rule.

The factors of trace-equivalent words are exactly the *convex* subsets of the circuit DAG (a set is convex iff it is an interval of some topological order). V2 of [1] applies $\mathcal{R}(8, 5)$ to random contiguous factors and explores the trace class by random commuting swaps, which amounts to a stochastic enumeration of convex windows. We enumerate convex windows directly instead.

## 3 The optimiser

**Databases.** For each device we fix a discrete grid of native gates on a 3-qubit local topology (all pairs for ions, the line 0–1–2 for heavy-hex). Heron: $\mathrm{RZ} \in \{\pm\frac{\pi}{4}, \pm\frac{\pi}{2}, \pi\}$, SX, $X$, CZ (23 gates). Ion: RX, RY, RZ on the same angles and $\mathrm{RXX}(\pm\frac{\pi}{4}, \pm\frac{\pi}{2})$ (57 gates). We enumerate words breadth-first and store, for each new class, a 64-bit hash of its canonical form (Table 1), minimised over the topology's automorphisms. The deepest level stores keys only. Every hit is re-checked numerically before use.

| device | topology | notion | depth | classes | build time |
|---|---|---|---|---|---|
| Heron | line | phase | 9 | 6.31M | 6 min |
| Heron | line | output-aware | 10 | 0.34M | <1 min |
| Heron | line | input-aware | 14 | 1.41M | <1 min |
| ion | triangle | phase | 6 | 28.4M | 38 min |
| ion | triangle | output-aware | 6 | 7.36M | 27 min |
| ion | triangle | input-aware | 7 | 31.6M | 23 min |
| paper ion ($\pm\frac{\pi}{2}$) | triangle | phase | 9 | 2.93M | 7 min |
| paper IBM ($\pm\frac{\pi}{2}$) | triangle | phase | 10 | 1.84M | 2 min |

Table 2: Hashed databases. The paper's databases were capped at 9–10 levels. They are *not* told that the group is finite or Clifford.

**Lookup beyond the radius.** If a window $U$ is not in the database, meet-in-the-middle tries $U = R\,A$ with $A$ in a stored full ball of radius 2. This extends the reach by two gates at the cost of one hash lookup per element of that ball.

**Continuous angles.** Circuits from real compilers use arbitrary angles, which no finite grid contains. Windows on at most two *coupled* wires are therefore also resynthesised analytically. On Heron we use KAK into RZ/SX/$X$/CZ with the minimal number of CZ. On ions, a Weyl decomposition yields one RXX($\theta$) per non-zero interaction coefficient. Single-qubit runs use Euler decompositions.

**Windows.** From each position $i$, a window grows by collecting later gates on a set of $\leq 3$ wires that commute (exactly, tolerance $10^{-10}$) past the gates they jump over. The result is a convex set. On Heron the wires must induce a connected subgraph, so every replacement is legal without routing. Every prefix of the window is tried, and the replacement with the largest cost saving is applied. Sweeps alternate with random exact-commutation shuffles until $p$ consecutive shuffles bring no gain ($p = 4$ on the hardware suite; $p = 8$ with 4 restarts on the paper's benchmarks).

**Global phase.** Inside windows the optimiser matches up to phase. It records each replacement's phase, snapped to the set the device can realise exactly ($\pi/4$ multiples on Heron, $\pi/2$ on ions). It then repairs the accumulated phase at the end with the cheapest gadget, or by absorbing $\pm i$ into an existing RXX on ions. Output is therefore equal *including* global phase. Qiskit and TKET store the phase as metadata instead, so we also report that convention.

**Guarantee.** The optimiser never returns a circuit of higher cost than its input: if repair would exceed the savings, it returns the input.

## 4 Theory for discrete gate sets

The following results concern the alphabets of [1]: $\Gamma_{\text{ion}} = \{\text{RX}, \text{RY}, \text{RZ}(\pm\frac{\pi}{2})\} \cup \{\text{RXX}(\pm\frac{\pi}{2})\}$ and $\Gamma_{\text{ibm}} = \{\text{RX}, \text{RZ}(\pm\frac{\pi}{2})\} \cup \{\text{CZ}\}$, modulo phase.

**Lemma 3** (The alphabets generate the Clifford group)**.** *$\langle\Gamma\rangle = \mathcal{C}_n/U(1)$, of order $2^{n^2+2n}\prod_{j=1}^{n}(4^j-1)$. Verified by enumeration for $n \leq 3$ (24; 11,520; 92,897,280). The compute graph of depth $d$ in [1] is the ball $B(d)$ of $\text{Cay}(G, \Gamma)$. All 22 entries of [1, Table I] are reproduced exactly.*

Write $|g|$ for the word length, $S(r)$ for the sphere, and $D$ for the diameter.

### 4.1 Exactness threshold and cascade

**Theorem 4** (Local = global above the diameter)**.** *Let $G$ be finite with diameter $D$. If $w \geq D+1$ and $r \geq D$, every $(w,r)$-reduced word is a geodesic.*

*Proof.* Let $u$ be $(w,r)$-reduced. If $|u| > D$, its first $D+1$ letters form a factor $f$ with $|\mathrm{ev}(f)| \leq D \leq r$ and $|f| \leq w$, so a rule applies, a contradiction. Hence $|u| \leq D < w$, so $u$ is itself a factor of length $\leq w$, and if it were not geodesic the rule $u \to v$ with $v$ geodesic would apply. □

**Proposition 5** (Cascade)**.** *Let $\ell(w)$ be the length of the fixed point of an exhaustive $\mathcal{R}(w,\infty)$ optimiser. If $\ell(w) \leq w$ at any time, the next sweep returns a geodesic.*

So the relevant window is $w^\star = \min\{w : \ell(w) \leq w\}$, which we measure: $w^\star \approx 12$ (ion trap) and 16 (IBM-style) for 3-qubit circuits, well below $D+1 = 18$. For continuous gate sets $G$ is infinite, $D = \infty$, and no finite window is exact. This is why ground truth must come from the discrete case.

### 4.2 Gate-count parity

**Theorem 6** (Parity)**.** *(a) For $n \in \{1,2\}$ and both alphabets, $\mathrm{Cay}(\mathcal{C}_n, \Gamma)$ is bipartite, so every equivalent rewrite removes an even number of gates. (b) For $n \geq 3$ and any generating set of $\mathcal{C}_n/U(1)$ that contains, for each qubit, a gate squaring to $X_i$ and one squaring to $Z_i$, the Cayley graph is not bipartite.*

*Proof.* A Cayley graph is bipartite iff some homomorphism $\chi : G \to \{\pm 1\}$ has $\chi(\Gamma) = -1$. (a) For $n = 1$, $\mathcal{C}_1/U(1) \cong S_4$, every $\pm\frac{\pi}{2}$ rotation is a 4-cycle, and $\chi = \mathrm{sgn}$. For $n = 2$ we verify it exhaustively. (b) The Pauli group is normal with quotient $\mathrm{Sp}(2n,2)$. Since $\chi(X_i) = \chi(\gamma)^2 = 1$ (and likewise $Z_i$), $\chi$ factors through $\mathrm{Sp}(2n,2)$, which is simple and non-abelian for $n \geq 3$. Hence $\chi \equiv 1$. □

The shortest odd relation for $n = 3$ has length $\rho = 11$ (ion trap) and 9 (IBM-style). A parity change therefore needs a window of at least $\lceil \rho/2 \rceil = 6$ gates, and Fig. 4 of [1], a $6 \to 3$ replacement on two qubits, cannot come from the stated gate set.

### 4.3 What random-circuit benchmarks measure

**Theorem 7** (Benchmarks mix)**.** *Let $\Gamma = \Gamma^{-1}$, $\mu_L$ the law of $\mathrm{ev}(u)$ for $u$ uniform in $\Gamma^L$, and $\beta < 1$ the largest non-trivial eigenvalue modulus of the random walk (excluding $-1$ if bipartite). Then $\|\mu_L - \pi_L\|_{TV} \leq \frac{1}{2}\sqrt{|G|}\,\beta^L$, with $\pi_L$ uniform on $G$ (or on the parity class). Hence $\mathbb{E}|\mathrm{ev}(u)| \to \frac{1}{|G|}\sum_d d\,|S(d)|$.*

This is the standard $L^2$ bound [19]. For $n = 2$ (ion trap) $\beta = 0.937$. Consequently a random length-300 benchmark has a *constant* optimal answer: 10.72 gates (3 qubits, ion trap) and 12.34 (IBM-style), independent of the input length beyond $L \approx 100$.

### 4.4 Lower bounds for every method

**Proposition 8** (Counting bound)**.** *For $g$ uniform in $G$, $\Pr[|g| \leq d] \leq |B(d)|/|G|$, and $|S(a+b)| \leq |S(a)||S(b)|$. Exact spheres up to depth $k$ thus give lower bounds on $\mathbb{E}|g|$ for every method.*

At $n = 4$ ($|G| = 1.21 \times 10^{13}$), spheres to depth 6 give $\mathbb{E}|g| \geq 11.1$ (ion trap) and $\geq 12.9$ (IBM-style). At $n = 3$ the same bound gives 7.9, against a true value of 10.72.

# 5 Exact ground truth

A Clifford modulo phase is determined by its signed tableau. For $n \leq 3$ we pack it into 64 bits, apply gates through lookup tables, and run BFS as vectorised gathers. Membership is exact.

| alphabet | $n$ | $\lvert\Gamma\rvert$ | $\lvert G\rvert$ | diameter | mean $\lvert g\rvert$ (uniform) |
|---|---|---|---|---|---|
| ion trap | 2 | 14 | 11,520 | 10 | 5.89 |
| IBM-style | 2 | 9 | 11,520 | 11 | 6.92 |
| ion trap | 3 | 24 | 92,897,280 | 17 | 10.72 |
| IBM-style | 3 | 15 | 92,897,280 | 19 | 12.34 |

Table 3: Exact Cayley-graph data (BFS: 25 min, ≈6 GB for $n = 3$).

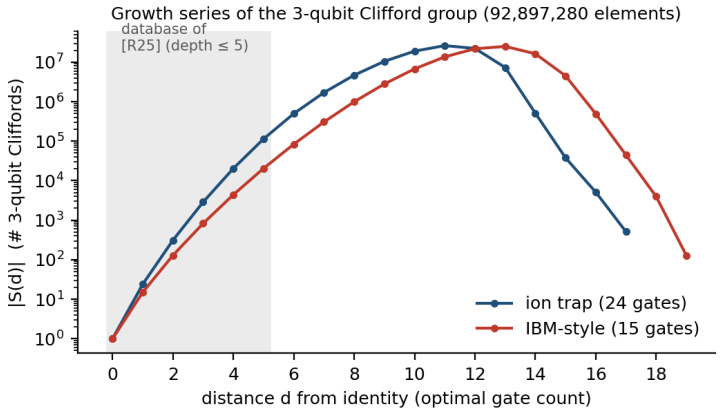



Figure 1: Growth series of the 3-qubit Clifford group. The radius-5 database of [1] (shaded) covers 0.15% of the group (ion trap). Almost all elements lie at distance 10–14.

For $n = 3$, the numbers of elements needing exactly $0, 1, \ldots, 6$ entangling gates are 13,824; 373,248; 5,971,968; 38,486,016; 42,052,608; 5,971,968; and 27,648 (mean 3.51). By local equivalence of RXX($\frac{\pi}{2}$) and CZ, the distribution is the same for both alphabets.

# 6 Experiments

**Verification.** Every output is checked against its input. Exact mode: $\|(U - V)\psi\|$ on random states (statevector on the active qubits). State-aware mode: the total-variation distance (TVD) of the output distributions on $|0^n\rangle$. Approximate mode: the measured error is compared with the tracked bound. Every output is also checked to be native and to use only coupled pairs.

## 6.1 The paper's benchmarks, with ground truth

We run the general optimiser with the paper's semantics (up to phase, gate count, $\pm\frac{\pi}{2}$ alphabets, all-to-all) on the paper's own settings and seeds. These are Tables III/IV (3 qubits, 300 gates, 20 circuits each), the journal's 4-qubit statistics (100 circuits), and Table V (15 qubits, 500 gates, 10 circuits). "Original" is our faithful port of the authors' V2, which reproduces the paper's single-circuit 3-qubit result (47.1 vs. 45).

| setting | Qiskit L3 | original [1] | Clifford CW-sem | **general** | optimum (or lower bound) | general at optimum |
|---|---|---|---|---|---|---|
| T3: ion, 3q | 127.2 | 47.1 | – | **10.8** | 10.4 | 17/20 |
| T4: IBM-style, 3q | 104.0 | 36.8 | – | **12.5** | 12.2 | 18/20 |
| journal: ion, 4q | 164.9 | 78.8* | 45.6 | **46.1** | ≥11.1 | – |
| journal: IBM-style, 4q | 140.9 | 76.0* | 49.3 | **51.4** | ≥12.9 | – |
| T5: IBM-style, 15q | 332.1 | 335.9 | 324.6 | **327.7** | – | – |

Table 4: Mean gate counts. *Original and Qiskit L3 at 4 qubits: first 30 circuits. "Clifford CW-sem" uses adaptive commutation-aware windows with the exact Clifford tables (Appendix A); the general engine does not use them. Time per circuit: 2.6 s (T3) to 8.9 s (T5). All 250/250 outputs verified and legal.

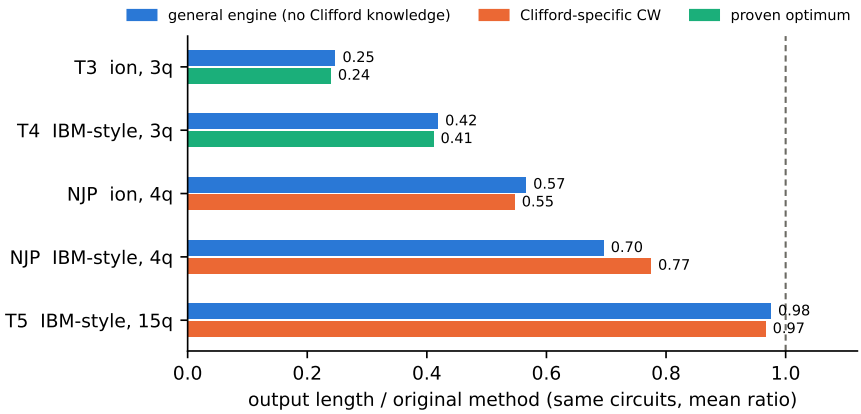



Figure 2: Output length relative to the original method on the same circuits (mean of per-circuit ratios).

**Findings.** (1) Without knowing that the group is finite or Clifford, the general engine is within 0.4 gates of the proven optimum on average and exactly optimal on 35 of 40 three-qubit circuits. The original method is 4.5× / 3.0× above the optimum. (2) At four qubits it is close to the Clifford-specific optimiser (ion: 46.1 vs. 45.6; IBM-style: 51.4 vs. 49.3), so knowing the group structure is not what makes the difference. Large databases and convex windows are. (3) At 15 qubits all methods, Qiskit L3 included, lie within 4% of one another. Here the windows see only a small part of a highly entangled random circuit, and the input is already close to a local optimum.

## 6.2 Algorithms on hardware models

**Suite.** We use 60 circuits of 2–12 qubits and up to 1008 gates: Nam/VOQC arithmetic [4] (Toffoli/Barenco ladders, adders, modular multiplication, GF($2^4$) multiplication, product formulas, QFT8), QASMBench [14] (small and medium circuits; circuits with classical feed-forward and four that do not parse are skipped), QFT, QAOA, Heisenberg Trotter, EfficientSU2, a Steane encoder, and random circuits. Each circuit is compiled for the device by Qiskit at level 1 (L1) and level 3 (L3) with the same coupling map and seed. Pipelines: TKET `FullPeepholeOptimise` on L1 followed by a rebase; ours on L1; ours on L3.

| | IBM Heron | | | trapped ion | | |
|---|---|---|---|---|---|---|
| pipeline (relative to Qiskit L3) | 2q gates | all gates | ESP | 2q gates | all gates | ESP |
| TKET (FullPeepholeOptimise on L1) | ×1.02 | ×1.71 | ×1.16 | ×1.27 | ×1.79 | ×0.992 |
| ours, exact, from L1 | ×1.00 | ×0.84 | ×1.21 | ×0.95 | ×0.76 | ×1.002 |
| ours, exact, from L3 | ×0.98 | ×0.87 | ×1.03 | ×1.00 | ×1.00 | ×1.001 |
| ours, approximate ($\varepsilon = 10^{-3}$), from L1 | ×1.01 | ×0.84 | ×1.21 | ×0.95 | ×0.75 | ×1.002 |
| ours, state/measurement-aware, from L1 | ×0.77 | ×0.61 | ×1.28 | ×0.62 | ×0.50 | ×1.010 |

Table 5: 58 circuits per device. Count columns are ratios of summed counts; ESP is the geometric mean of per-circuit ratios. Nam pf1_10/pf2_10 are excluded because Qiskit L3 drops their $\sim 10^{-7}$ rad rotations (an approximation, not an exact rewrite): L3 reports 0 and 0 two-qubit gates, against 139 and 298 for our exact output.

**Findings.** (1) *Stacked on Qiskit L3, the optimiser never adds a two-qubit gate.* Heron: 9 circuits better, 49 equal. Ion: 1 better, 57 equal. ESP never decreases (0 of 58 on either device). On Heron it removes 13% of all gates: SX/$X$ pulses fall from 7218 to 6139, virtual RZ from 5297 to 4436, and CZ from 2889 to 2830. On ions the total count can *rise* (28 circuits) because the calibrated objective replaces RX/RY pulses with virtual RZ gates (pulses 3824 → 2422, RZ 2223 → 3622). This is intended, but it means "never worse" holds for the objective and the two-qubit count, not for the total. (2) *From L1, ours beats L3 on Heron* in all gates (×0.84) and ESP (×1.21), and matches it in two-qubit count. The ESP gain is not all ours: Qiskit's L1 layout already has ×1.11 the ESP of its L3 layout (neither is noise-aware, and our optimiser never changes the layout), and our rewriting adds ×1.09 on top. On ions it removes 5% of the MS gates that L3 leaves, through Weyl resynthesis with arbitrary-angle RXX. (3) *TKET* keeps its two-qubit counts close to L3 but uses 1.7–1.8× as many gates after rebasing. (4) *State/measurement-aware* rewriting gives the largest gains (×0.77 / ×0.62 two-qubit gates). Part of this gain is trivial: 5 (Heron) and 7 (ion) circuits are reversible logic on $|0\cdots0\rangle$ whose output is a fixed basis state, and they reduce to the empty circuit. Such programs are a property of the benchmark suite, so this mode should be read per circuit, not as an average. (5) The approximate mode with $\varepsilon = 10^{-3}$ per replacement gives almost nothing beyond exact mode on these circuits, which contain few near-identity blocks. Measured errors (max $6.4 \cdot 10^{-4}$) are always within the tracked bound. (6) Run time: median 10.4 s (Heron) and 3.0 s (ion) per circuit and configuration, maximum 180 s. This is Python and single-threaded. Qiskit L3 takes well under a second.

### 6.3 Choosing the objective

We ran the same optimiser with $c_{\text{count}}$, $c_{w2}$ and $c_{\text{cal}}$ and simulated each output with noise. Heron uses the calibrated FakeTorino gate errors of the actual physical qubits and pairs, remapped onto the active register. Ions use depolarizing noise at the model rates. We report the exact TVD between the ideal and noisy output distributions, computed from density matrices, on the 49 circuits with at most 8 active qubits.

| mean TVD | Qiskit L1 | Qiskit L3 | ours, count | ours, $w2$ | ours, calibrated |
|---|---|---|---|---|---|
| Heron | 0.115 | 0.137 | 0.096 | 0.088 | **0.088** |
| ion | 0.0157 | 0.0091 | 0.0092 | 0.0084 | **0.0084** |

Table 6: Noisy-simulation error by objective. Calibrated beats L3 on 39/49 (Heron) and 29/49 (ion) circuits.

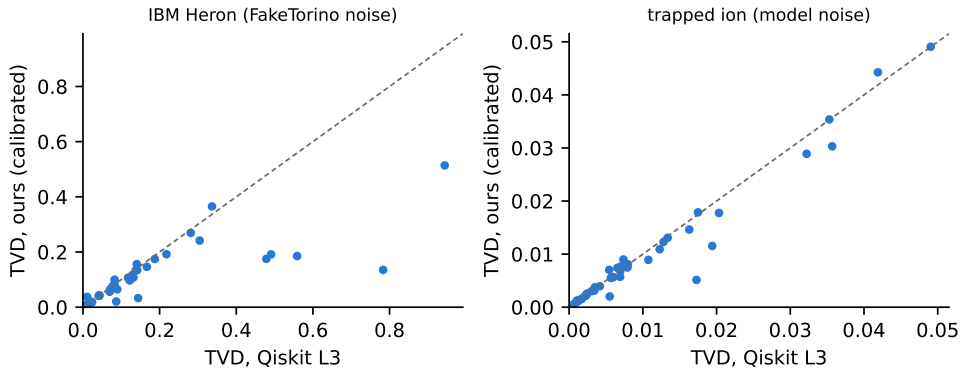



Figure 3: Per-circuit noisy TVD: Qiskit L3 against ours (calibrated, from L1). Points below the diagonal favour ours.

On both devices the calibrated cost is at least as good as the alternatives, and on Heron it is clearly better than plain count. The two-qubit-weighted count is almost as good, so on devices without calibration data it is a reasonable proxy. We use $c_{\text{cal}}$ as the default. The ion comparison is conservative for our method: the simulation charges RZ a depolarizing error of $3 \cdot 10^{-5}$, while the cost model treats it as virtual.

## 7 Limitations and threats to validity

- **Verification.** Verification is numerical (random input states), not symbolic. On Heron, the routed layout of Nam pf2_10 spans 20 physical qubits, more than the main run's statevector cap of 16, so its five L1-based outputs were verified in a separate re-run (same totals; exact error $\leq 3.3 \cdot 10^{-11}$, approximate error $1.3 \cdot 10^{-4}$ within its bound $7.1 \cdot 10^{-4}$). All other 360 (Heron) and 360 (ion) outputs, and the 250 paper-benchmark outputs, were verified in the run itself (exact error $\leq 8.2 \cdot 10^{-11}$, state-aware TVD $\leq 6 \cdot 10^{-12}$).
- **Noise models, not hardware.** ESP and TVD come from calibration snapshots and simulators. The Heron model contains gate errors only: no readout, idle (decoherence during waiting), or crosstalk errors. The ion parameters are assumptions.
- **Global phase.** Our outputs are exact including phase. On Heron this costs 240 single-qubit gates over the suite. With phase as metadata, as Qiskit and TKET store it, our all-gate ratio from L1 improves from $\times 0.838$ to $\times 0.825$.
- **Baselines.** BQSKit was not available in our environment, and Quartz/QUESO were not run. Qiskit and TKET are used at their defaults with one seed. Layout and routing are Qiskit's, and our optimiser does not change the layout.
- **Scale.** Windows have at most 3 qubits, and the code is single-threaded Python. Circuits beyond $\sim$1,000 gates were not tested.
- **Feed-forward.** Circuits with mid-circuit measurement and classical control are out of scope.

## 8 Weaknesses of the original paper

1. **Missing structure.** The gate sets are Clifford, so the search space is a finite group with exact fingerprints and a complete theory [11, 12, 10]. None of it is used. The paper uses floating-point comparisons with inconsistent tolerances ($10^{-5}$ in the text, $10^{-3}/10^{-4}$ in the code) and an $O(N)$ linear scan where an $O(1)$ key exists. It then adds a classifier (V3) to save time on that lookup.
2. **No notion of optimality.** The paper compares only with Qiskit. Its 3-qubit result is $4.5\times$ the optimum, which can be computed in minutes.
3. **Equivalence notion.** Up-to-phase replacement is unsound under control, and the paper does not discuss it. Neither the state/measurement context nor approximation is used, and no error

is tracked.

4. **Hardware model.** Discrete $\pm\pi/2$ angles, all-to-all connectivity, and total gate count as the objective. None of these matches IBM (RZ($\theta$), heavy-hex) or ion (continuous MS) hardware.
5. **Evaluation protocol.** Random circuits mix (Thm. 7), so "from 300 gates" measures the input length. Tables III–V use one circuit each. Times compare MATLAB with compiled transpilers.
6. **Search.** Windows are contiguous. Commutation is explored by random swaps, with no convergence criterion. The radius-5 database misses windows whose optimum is 6–7 gates.
7. **V3.** Features, forest and threshold are unreported. Reproduced recall is 30% at the default threshold.
8. **Hardware evidence.** One 2-qubit circuit, one run per chip, no error bars. In our reproduction the effect disappears at Qiskit level 3.
9. **Reproducibility.** The released code contains only V2 and has visible bugs (`simplereduce` skips the last pair; the database excludes the identity). The database used for Tables III–V is not stated.

## 9 Open problems

1. **Local-to-global theory.** Characterise $w^\star(G, \Gamma)$. Prove the cascade threshold for random inputs from the growth series, and relate it to Dehn presentations and $k$-local geodesics [20, 21].
2. **Ground truth beyond Clifford.** With $\pm\pi/4$ angles the group is infinite (Clifford+T). Exact normal forms [15, 16] could still give certified optima for small $T$-counts.
3. **Larger windows.** 4-qubit databases via symmetry reduction, or a learned admissible heuristic for A$^*$ trained on exact tables.
4. **Layout-aware rewriting.** Joint optimisation of layout and local rewriting, using calibrated costs.
5. **Real hardware.** Runs with error bars at fixed transpiler settings, reporting two-qubit counts and ESP.

## 10 Conclusion

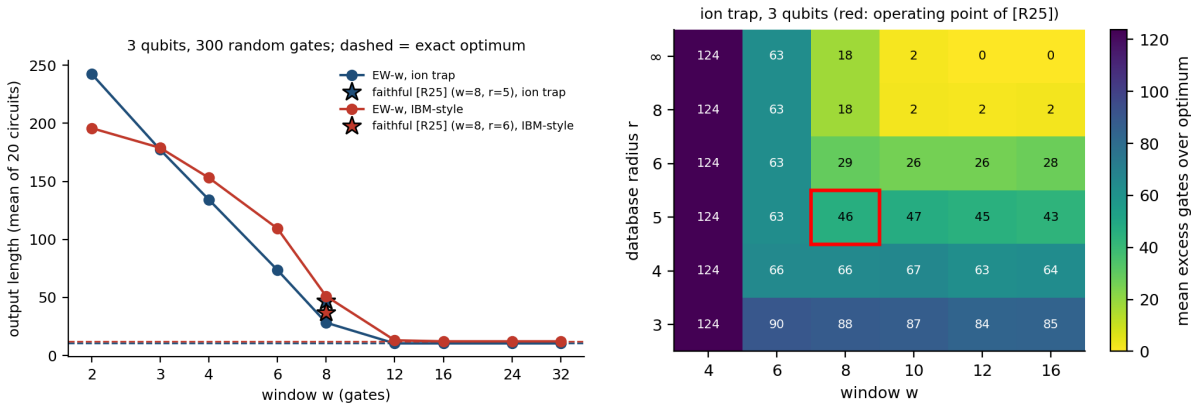
Local rewriting becomes reliable when "equivalent" is made precise. Each notion is a group action, the database stores its orbits, and soundness and error accumulation follow. A general optimiser built this way, without using the structure of any gate set, reaches proven optima on the original paper's 3-qubit benchmarks, at a quarter of the original method's output length. It also improves Qiskit level-3 output on real algorithms for Heron and ion models without ever adding a two-qubit gate. The Clifford ground truth is what makes such claims checkable.

## A Clifford-specific results

For the paper's alphabets, the exact tables of §5 give an *exact-window* optimiser EW-$w$ (all windows of length $\leq w$ on $\leq 3$ wires, replaced by optimal words) and its DAG-convex version CW-$w$.

| 3 qubits, 300 gates | optimum | original [1] | EW-8 | EW-16 | Qiskit L3 | paper (1 circuit) |
|---|---|---|---|---|---|---|
| ion trap (20) | 10.4 | 47.1 | 28.3 | 10.4 | 127.2 | 45 |
| IBM-style (20) | 12.2 | 36.8 | 51.1 | 12.2 | 104.0 | 72 |
| time per circuit (ion) | – | 51 s | 0.54 s | 0.24 s | <0.1 s | – |

Table 7: Mean gate counts in the setting of Tables III/IV of [1].




(a) (b)

Figure 4: (a) Fixed-point length of EW-$w$ against $w$, with the cascade transition near $w^\star$. (b) Mean excess over the optimum by window $w$ and database radius $r$ (ion trap). The original operating point $(8, 5)$ is marked.

**Findings.** (1) On its own benchmark the original method finishes 4.5× (ion trap) and 3.0× (IBM-style) above the exact optimum. Our faithful port reproduces the paper's single-circuit number (47.1 vs. 45 gates), so the gap is a property of the method, not of our implementation. (2) EW-$w$ reaches the *proven* optimum on all 20 circuits once $w \geq 12$ (ion trap) or $w \geq 16$ (IBM-style), in a fraction of a second. This is the cascade of Prop. 5: the transition is sharp, between $w = 8$ and $w = 12$ (ion trap) or $w = 16$ (IBM-style). (3) The grid (Fig. 4b) shows that the *database radius*, not the window or the search strategy, is the binding constraint. At $r = 5$, enlarging the window from 8 to 16 lowers the mean excess only from 46 to 43 gates. At $w = 8$, raising $r$ from 5 to $\infty$ lowers it from 46 to 18, and for $w \geq 12$ with $r \geq 8$ the excess is at most 2 gates (ion trap; IBM-style: from 72 to $\leq 8$). Theorem 4 predicts both requirements: windows long enough *and* a database deep enough. The original method is limited by the second.

| 4 qubits, 300 gates, 100 circuits | lower bound | CW-24 | EW-16 | original | published [1] | Qiskit L3 | BQSKit L4 [1] |
|---|---|---|---|---|---|---|---|
| ion trap: total | ≥11.1 | 56.7 | 110.3 | 78.8 | 111 | 164.9 | 126 |
| ion trap: RXX | – | 27.6 | 44.2 | 33.3 | 43 | 47.3 | 28 |
| IBM-style: total | ≥12.9 | 59.1 | 130.9 | 76.0 | 107 | 140.9 | 168 |
| IBM-style: CZ | – | 27.1 | 47.1 | 32.4 | 43 | 47.3 | 37 |

Table 8: Journal benchmark. Original and Qiskit: first 30 circuits. Published and BQSKit numbers are from the journal version of [1].

**Findings.** (1) *Convexity matters more than anything else at $n \geq 4$.* With the same exact 3-qubit oracle, contiguous windows (EW-16) give 110.3 (ion trap) and 130.9 (IBM-style) gates, no better than the published method. With DAG-convex windows (CW-24) the same oracle gives 56.7 and 59.1 gates, i.e. 49% and 45% fewer than the published results, in about 5.6 s per circuit. (2) Two-qubit counts drop from 43 (published) to 27.6 RXX and 27.1 CZ, on par with BQSKit L4 for the ion trap (28) and below it for the IBM-style set (37), although CW optimises total length, not two-qubit count. (3) The certified lower bounds (11.1, 12.9) leave a gap of at most a factor $\approx 5$. Closing it, either by tightening the bound (deeper spheres, meet-in-the-middle) or by better upper bounds (4-qubit oracles via symmetry reduction), is an open problem that our framework makes precise.

**Adaptive windows.** CW-sem replaces CW-24's fixed 3-wire windows by windows that grow while gates commute into them. With the same exact oracle it reaches 45.6 (ion trap) and 49.3 (IBM-style) gates on the same 100 circuits; these are the numbers used in Table 4.

# References


[1] B. Rosenhahn, T. J. Osborne, C. Hirche, Optimization driven quantum circuit reduction, arXiv:2502.14715 (2025); New J. Phys. **27**, 104509 (2025).

[2] A. K. Prasad, V. V. Shende, I. L. Markov, J. P. Hayes, K. N. Patel, Data structures and algorithms for simplifying reversible circuits, ACM JETC **2**(4), 277 (2006).

[3] O. Golubitsky, D. Maslov, A study of optimal 4-bit reversible Toffoli circuits and their synthesis, IEEE Trans. Comput. **61**(9), 1341 (2012).

[4] Y. Nam, N. J. Ross, Y. Su, A. M. Childs, D. Maslov, Automated optimization of large quantum circuits with continuous parameters, npj Quantum Inf. **4**, 23 (2018).

[5] R. Iten, R. Moyard, T. Metger, D. Sutter, S. Woerner, Exact and practical pattern matching for quantum circuit optimization, ACM Trans. Quantum Comput. **3**(1) (2022).

[6] M. Xu, Z. Li, O. Padon, S. Lin, J. Pointing, A. Hirth, H. Ma, J. Palsberg, A. Aiken, U. A. Acar, Z. Jia, Quartz: superoptimization of quantum circuits, PLDI (2022).

[7] A. Xu, A. Molavi, L. Pick, S. Tannu, A. Albarghouthi, Synthesizing quantum-circuit optimizers, PLDI (2023).

[8] Z. Li et al., Quarl: a learning-based quantum circuit optimizer, OOPSLA (2024).

[9] S. Bravyi, R. Shaydulin, S. Hu, D. Maslov, Clifford circuit optimization with templates and symbolic Pauli gates, Quantum **5**, 580 (2021).

[10] S. Bravyi, J. A. Latone, D. Maslov, 6-qubit optimal Clifford circuits, npj Quantum Inf. **8**, 79 (2022).

[11] S. Aaronson, D. Gottesman, Improved simulation of stabilizer circuits, Phys. Rev. A **70**, 052328 (2004).

[12] P. Selinger, Generators and relations for $n$-qubit Clifford operators, Log. Methods Comput. Sci. **11**(2) (2015).

[13] S. Nishio, Y. Pan, T. Satoh, H. Amano, R. Van Meter, Extracting success from IBM's 20-qubit machines using error-aware compilation, ACM JETC **16**(3) (2020).

[14] A. Li, S. Stein, S. Krishnamoorthy, J. Ang, QASMBench: a low-level quantum benchmark suite for NISQ evaluation and simulation, ACM Trans. Quantum Comput. **4**(2) (2023).

[15] B. Giles, P. Selinger, Exact synthesis of multiqubit Clifford+T circuits, Phys. Rev. A **87**, 032332 (2013).

[16] V. Kliuchnikov, D. Maslov, M. Mosca, Fast and efficient exact synthesis of single-qubit unitaries generated by Clifford and T gates, Quantum Inf. Comput. **13**, 607 (2013).

[17] P. Cartier, D. Foata, Problèmes combinatoires de commutation et réarrangements, LNM 85, Springer (1969).

[18] V. Diekert, G. Rozenberg (eds.), The Book of Traces, World Scientific (1995).

[19] P. Diaconis, Group Representations in Probability and Statistics, IMS Lecture Notes 11 (1988).

[20] M. Gromov, Hyperbolic groups, in Essays in Group Theory, MSRI Publ. 8, Springer (1987).

[21] D. B. A. Epstein et al., Word Processing in Groups, Jones and Bartlett (1992).